Cristian Mesquida

Postdoctoral Researcher

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Employment

Postdoctoral researcher — *Eindhoven, The Netherlands*

Eindhoven University of Technology — *2025-Present*

- Contributed to the development and validation of Metacheck, an R package for improving research quality and extracting data from published articles.

PhD candidate — *Eindhoven, The Netherlands*

Eindhoven University of Technology — *2021-2025*

- My PhD thesis examined the factors hindering replicability and reproducibility in sports and exercise science.

Education

PhD — *Eindhoven, The Netherlands*

Eindhoven University of Technology — *2021-2025*

- My PhD thesis focused on the factors that hinder replicability and reproducibility of research in sports and exercise science.

MSc. Integrative Physiology — *Barcelona, Spain*

University of Barcelona — *2017-2019*

- My master thesis focused on the effect of two training protocols on maxymal oxygen uptake in highly-trained cyclists.

MSc. Research in Sports and Physical Activity — *Granada, Spain*

University of Granada — *2015-2017*

- My master thesis focused on the performance determinants of swimming sprint performance.

BSc. Sports and Exercise Science — *Vic, Spain*

University of Vic — *2009-2013*

- My bachelor end project focused on developing a training programe to reduce the injury risk of hamstrings in football players.

Publications

1. Lakens, D., Mesquida, C., Xavier-Quintais, G., Rasti, S., Toffalini, E., & Altoè, G. (2026). Rethinking Type S and Type M Errors. *Advances in Methods and Practices in Psychological Science*, *9*(2), 25152459261432530. https://doi.org/10.1177/25152459261432530
2. Ditroilo, M., Mesquida, C., Abt, G., & Lakens, D. (2025). Exploratory research in sport and exercise science: Perceptions, challenges, and recommendations. *Journal of Sports Sciences*, 1–13. https://doi.org/10.1080/02640414.2025.2486871
3. Murphy, J., Caldwell, A. R., Mesquida, C., Ladell, A. J. M., Encarnación-Martínez, A., Tual, A., Denys, A., Cameron, B., Van Hooren, B., Parr, B., DeLucia, B., Mason, B. R. J., Clark, B., Egan, B., Brown, C., Ade, C., Sforza, C., Taber, C. B., Kirk, C., … Warne, J. P. (2025). Estimating the Replicability of Sports and Exercise Science Research. *Sports Medicine*. https://doi.org/10.1007/s40279-025-02201-w
4. Lakens, D., Mesquida, C., Rasti, S., & Ditroilo, M. (2024). The benefits of preregistration and Registered Reports. *Evidence-Based Toxicology*, *2*(1), 2376046. https://doi.org/10.1080/2833373X.2024.2376046
5. Mesquida, C., & Lakens, D. (2024). Is the effect large enough to matter? Why exercise physiologists should interpret effect sizes meaningfully: A reply to Williams et al. (2023). *The Journal of Physiology*, *602*(1), 241–242. https://doi.org/10.1113/JP285901
6. Holgado, D., Mesquida, C., & Román-Caballero, R. (2023). Assessing the Evidential Value of Mental Fatigue and Exercise Research. *Sports Medicine*, *53*(12), 2293–2307. https://doi.org/10.1007/s40279-023-01926-w
7. Mesquida, C., Murphy, J., Lakens, D., & Warne, J. (2023). Publication bias, statistical power and reporting practices in the Journal of Sports Sciences: Potential barriers to replicability. *Journal of Sports Sciences*, *41*(16), 1507–1517. https://doi.org/10.1080/02640414.2023.2269357

8. Murphy, J., Mesquida, C., & Warne, J. (2023). A Survey on the Attitudes Towards and Perception of Reproducibility and Replicability in Sports and Exercise Science. *Communications in Kinesiology*, *1*(5). https://doi.org/10.51224/cik.2023.53
9. Bossi, A. H., Mesquida, C., Hopker, J., & Rønnestad, B. R. (2023). Adding Intermittent Vibration to Varied-intensity Work Intervals: No Extra Benefit. *International Journal of Sports Medicine*, *44*(2), 126–132. https://doi.org/10.1055/a-1812-7600
10. Mesquida, C., Murphy, J., Lakens, D., & Warne, J. (2022). Replication concerns in sports and exercise science: A narrative review of selected methodological issues in the field. *Royal Society Open Science*, *9*(12), 220946. https://doi.org/10.1098/rsos.220946
11. Murphy, J., Mesquida, C., Caldwell, A. R., Earp, B. D., & Warne, J. P. (2022). Proposal of a Selection Protocol for Replication of Studies in Sports and Exercise Science. *Sports Medicine*. https://doi.org/10.1007/s40279-022-01749-1
12. Etxebarria, N., Wright, J., Jeacocke, H., Mesquida, C., & Pyne, D. B. (2021). Running Your Best Triathlon Race. *International Journal of Sports Physiology and Performance*, *16*(5), 744–747. https://doi.org/10.1123/ijspp.2020-0838
13. Bossi, A. H., Mesquida, C., Passfield, L., Rønnestad, B. R., & Hopker, J. G. (2020). Optimizing Interval Training Through Power-Output Variation Within the Work Intervals. *International Journal of Sports Physiology and Performance*, *15*(7), 982–989. https://doi.org/10.1123/ijspp.2019-0260

Preprints

1. Mesquida, C., Warne, J., & Lakens, D. (2025). *A scoping review of the transparency, reporting practices and methodological rigor of meta-analyses published in Sports Medicine*. SportRxiv. https://doi.org/10.51224/SRXIV.576
2. Mesquida, C., Murphy, J., Warne, J., & Lakens, D. (2025). *Prevalence, reporting practices, and methodological quality of a priori power analyses in sports and exercise science research*. SportRxiv. https://doi.org/10.51224/SRXIV.575
3. Mesquida, C., Warne, J., & Lakens, D. (2025). *What is your hypothesis? On the importance of knowing your hypothesis before conducting a hypothesis test*. SportRxiv. https://doi.org/10.51224/SRXIV.577
4. Mesquida, C., Murphy, J., Warne, J., & Lakens, D. (2025). *On the replicability of sports and exercise science research: Assessing the prevalence of publication bias and studies with underpowered designs by a z-curve analysis*. SportRxiv. https://doi.org/10.51224/SRXIV.534

Papers in preparation

These are manuscripts with an existing draft that are nearing release as preprints.

1. DeBruine, L., Mesquida, C., & Lakens, D. (2026). *Introducing Metacheck as an Automated Tool for Improving Research Practices in Psychology*. https://github.com/scienceverse/ms_introduction_metacheck
2. Mesquida, C., DeBruine, L., & Lakens, D. (2026). *Improving the Transparency of Power Analyses Through Standardized Reporting*. https://github.com/scienceverse/ms_power_standards
3. Lakens, D., Mesquida, C., & DeBruine, L. (2026). *Best Practices for Machine Readable Data and Code Sharing in Psychology*. https://github.com/scienceverse/ms_FAIR_data_and_code
4. Warne, J., Murphy, J., Lakens, D., & Mesquida, C. (2026). *Statistical Reporting Practices, P-value Reproducibility and Data Sharing in 350 Sport and Exercise Science Studies: A Manual and Automated Audit*. https://github.com/cristianmesquida/reporting_practices

Ongoing Projects

These are ongoing research projects for which data collection is still underway or that are still at an early stage.

1. Nolte, S., Muprhy, J., Mesquida, C., & Rein, R. (2027). *Computational Reproducibility in Sports Science: A Registered Report Reanalysing Private Raw Data*.
2. Mesquida, C., Postelnicu, A., & Lakens, D. (2027). *Computational Reproducibility of Power Analyses in Large-Scale Replication Projects: Insights for Quality Control*.
3. Mesquida, C., Nuijten, M., Lakens, D., & Debruine, L. (2027). *Making Validation Mainstream in Automated Research Tools: Principles, Challenges, and FAIR Practices*.

Teaching Experience

My teaching portofolio can be found **here**.

Teacher of Advanced Research Methods and Research Ethics

Eindhoven University of Technology *2025-Present*

- Taught p-values and Bayes factors, power analysis, and equivalence testing.

Teacher of Behaveoral Research Methods 2: Dealing with data

Eindhoven University of Technology *2025-Present*

- Taught multiple regression, variable selection and power analysis.

Teacher of Metascience

Eindhoven University of Technology *2024-Present*

- Taught introduction to meta-analysis and bias-detection tests.

Teaching assistant of Advanced Research Methods and Research Ethics

Eindhoven University of Technology *2024*

- Assisted students during tutorial sessions with course assignments.

Teaching assistant of Advanced Data Analysis

Eindhoven University of Technology *2024*

- Taught R Markdown for reproducible reporting and data simulation for statistical analysis.

Teaching Assistant of Applied Exercise Physiology

Technological University of Dublin *2021*

- Taught human anatomy.

Supervision

Supervisor of Master End Project

Eindhoven University of Technology *2026*

- Co-supervised a project creating an automated tool that categorizs repository files based on the file name, index detected variables and their labels, and reconstruct the repository into an established standard.

Supervisor of Master End Project

Eindhoven University of Technology *2025*

- Co-supervised a project using automated search extraction to extract and classify hypotheses using large language models.

Supervisor of Bachelor End Project

Eindhoven University of Technology *2025*

- Co-supervised a meta-analysis project examining the impact of school sanitation on students' academic performance.

Supervisor of Master End Project

Eindhoven University of Technology *2024*

- Co-supervised the development of a tool to identify incorrect use of the standard error of measurement instead of the standard deviation in effect size calculations for meta-analyses.

Qualifications

Start to Supervise

The Netherlands

Eindhoven University of Technology *2026*

- Enabling Course for main supervisors.

National Triathlon Coach

Spain

Spanish Triathlon Federation *2018*

Talks

My talks and workshops can be found **here**.

Lunch Talk

German Sport University Cologne *2026*

- Discussed the main quality issues in power analyses reported in sports and exercise science research.

FORRT AI in Metascience

FORRT *2026*

- Discussed the lack of transparency in power analysis reporting in psychology and how Metacheck can help improve reporting practices.

ONH Metascience Symposium

Oslo Nye Høyskole *2026*

- Presented findings on the quality of power analyses for sample size justification in sports and exercise science.

Transparency, Technology and AI in Peer Review

Eindhoven University of Technology *2026*

- Presented Metacheck, an automated tool for assessing research transparency.

Workshop on conducting power analyses

Universitat de Vic *2026*

- Delivered a workshop on power analyses where PhD researchers were introducted to basic power analyses and simulation-based power analyses.

Metascience Series

Charles Sturt University *2025*

- Gave a talk on research waste in sports and exercise science.

Workshop on Metacheck

Open Research Summer School at King's College London *2025*

- Delivered a workshop on Metacheck, including a presentation and hands-on session.

Danish Reproducibility Network meetup

Danish Reproducibility Network *2025*

- Gave a talk on the benefits of preregistration and Registered Reports in hypothesis-testing research.

Lunch meetup for the Open Science Community

Eindhoven University of Technology *2025*

- Gave a talk on the benefits of preregistration and Registered Reports in hypothesis-testing research.

NLRN Pre-Symposium for Young Meta-Scientists

University Medical Center Groningen *2024*

- Presented findings on publication bias and statistical power in sports and exercise science.

Paul Meehl Graduate School PhD Day Conference

Eindhoven University of Technology *2024*

- Presented findings on the reporting and reproducibility of a priori power analyses in sports and exercise science.

Workshop on endurance performance

Setanta College *2021-2023*

- Conducted a workshop on the physiological determinants of endurance and team-sport performance, including lactate-threshold testing.

Workshop on 1080 Sprint resistance device

Kumming Hiageng Physical Training Base *2019*

- Conducted a workshop on using the 1080 Sprint to assess swimmers' power–velocity profiles and drag.

Grants

Replication studies

Dutch Research Council *2026*

- Applied as a co-applicant. The funded project aims to assess the quality and reproducibility of power analyses in large-scale replication projects (Project details at https://doi.org/10.61686/FSWVF35053).

IE&IS ESA Education Innovation Grant

Eindhoven University of Technology *2024*

- Applied as a co-applicant. The funded project aims to develop an Metacheck modules designed to assist master's students and supervisors in performing automatic checks to enforce best research practices.

Conference organization

Transparency, Technology and AI in peer review Conference *The Netherlands*

Eindhoven University of Technology *2026*

- I co-organized a one-day conference that brought together different stakeholders to discuss the role of transparency, technology, and AI in peer review (https://meta-eindhoven.github.io/events/)

Skills

R (advanced)

- Data wrangling, statistical analysis, simulation and reproducible materials using Quarto.

R (developing)

- Unit testing and package development.

Python (beginner)

- I am currently enrolled in a course to learn Python.

Git/Github ((intermediate)

- Version control, collaborative workflows and project management.

Statistical analysis

- Linear and non-linear regression, mixed-effects models, (multi-level) meta-analysis and power analysis.

Open science practices

- Preregistration, documentation and FAIR data and code.

Languages

Dutch

- Beginner (A1, currently learning)

English

- Advanced (C1)

Catalan

- Native

Spanish

- Native